PRACTICAL 11

Advanced CRUD Operations, Searching, Filtering, Soft Deletes, and CSV Export

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Program / Semester	| 3RD
Course / Subject	| WEB DEVELOPMENT FRAMEWORK
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Submission Date	| 31-10-2026

1. Aim

To implement an advanced student management interface with full CRUD functionality, searching, course-wise filtering, soft delete mechanisms, and CSV export capabilities using PHP and MySQL.

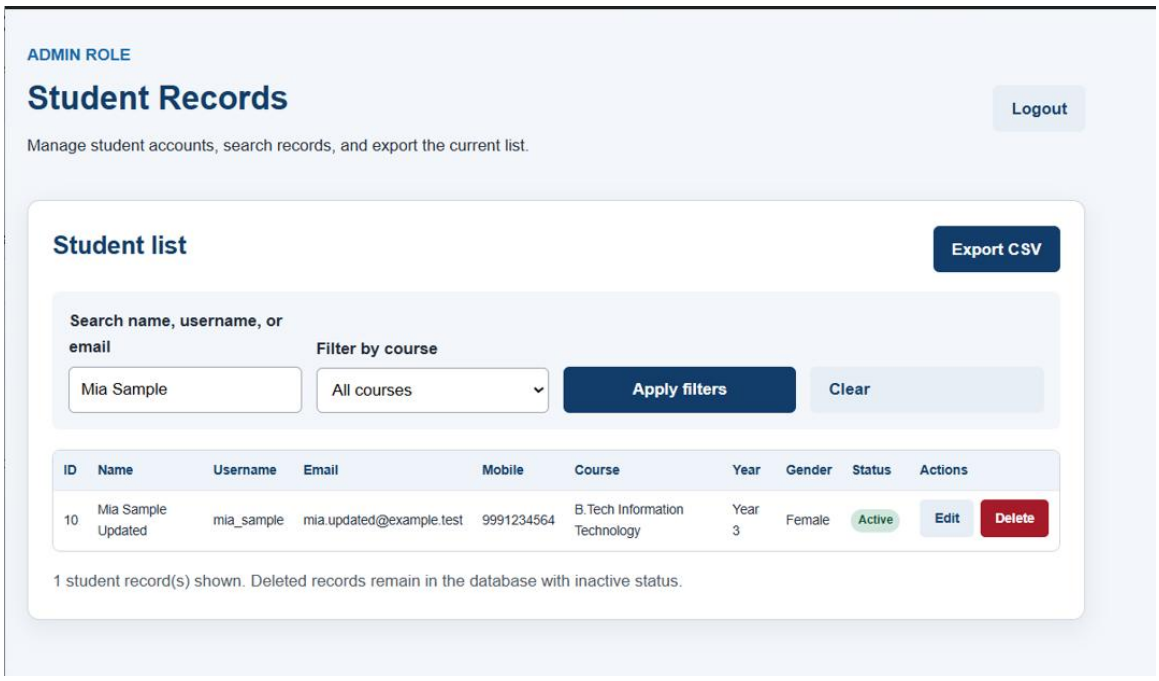
2. Practical Requirements

• CRUD Operations – Implement functionality to Create (add), Read (display in a table), Update, and Delete (soft delete) student records.
• Search & Filter – Add capabilities to search for specific students using keywords and filter the data dynamically by course.
• Status Management (Soft Delete) – Implement a soft delete feature where the student's status is updated to 'inactive' instead of permanently removing the row from the database.
• User Feedback – Display appropriate success or error messages for every operation (add, update, delete).
• Data Export – Create a script to generate and force-download a CSV file containing the currently queried student records.

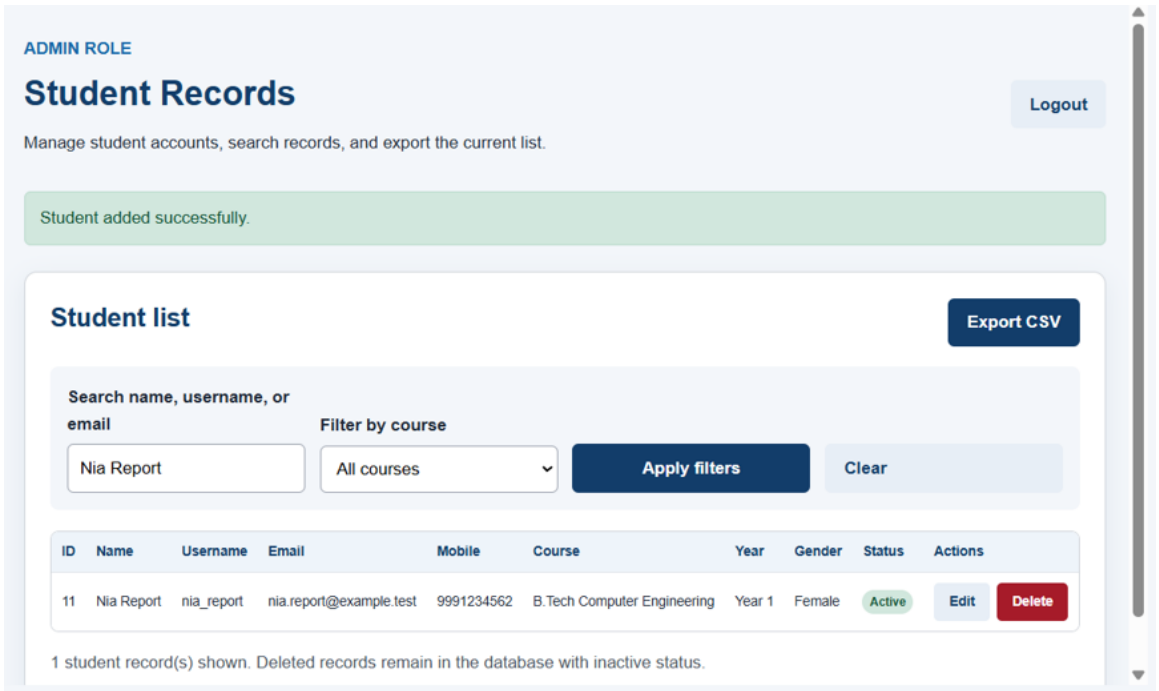
3. Required Screenshots / Demonstration

Insert the screenshots below in the same order as the practical instructions.

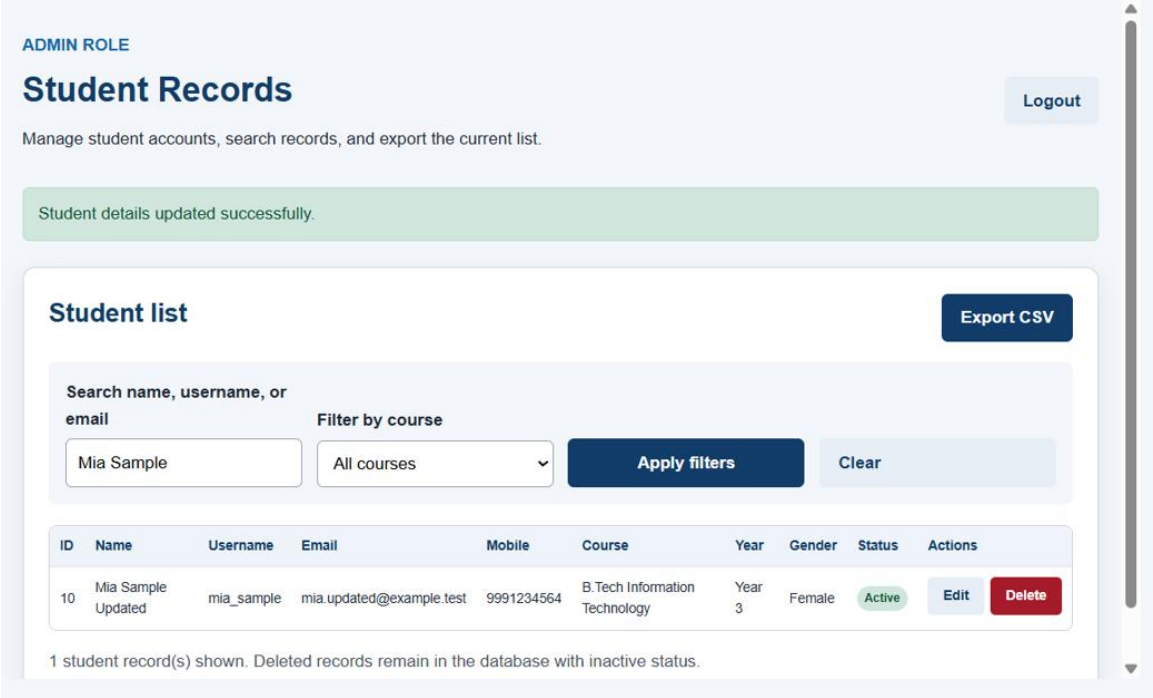
3.1 Student records displayed in a table



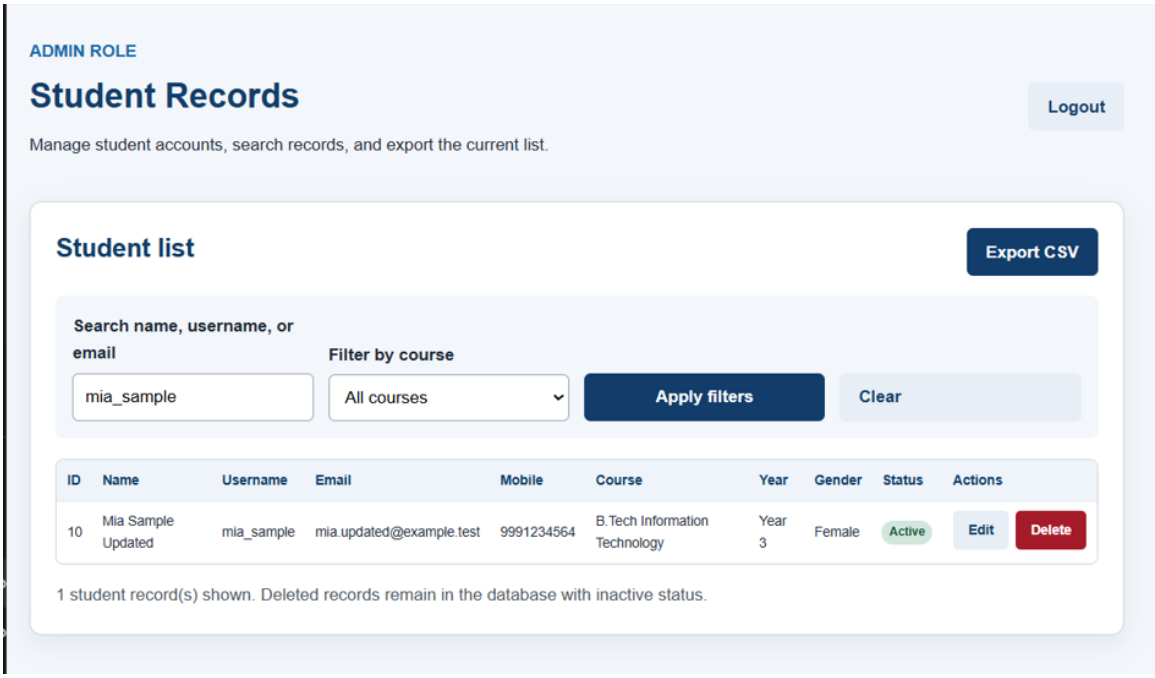
3.2 Successful addition of a new student



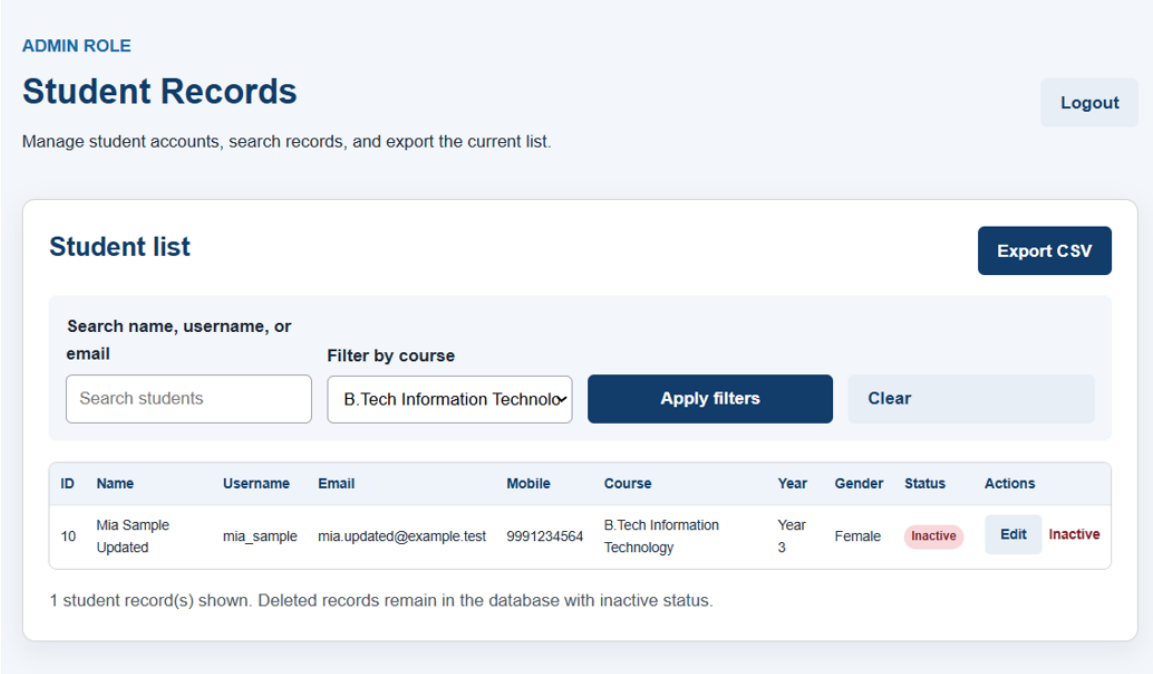
3.3 Successful update of student details



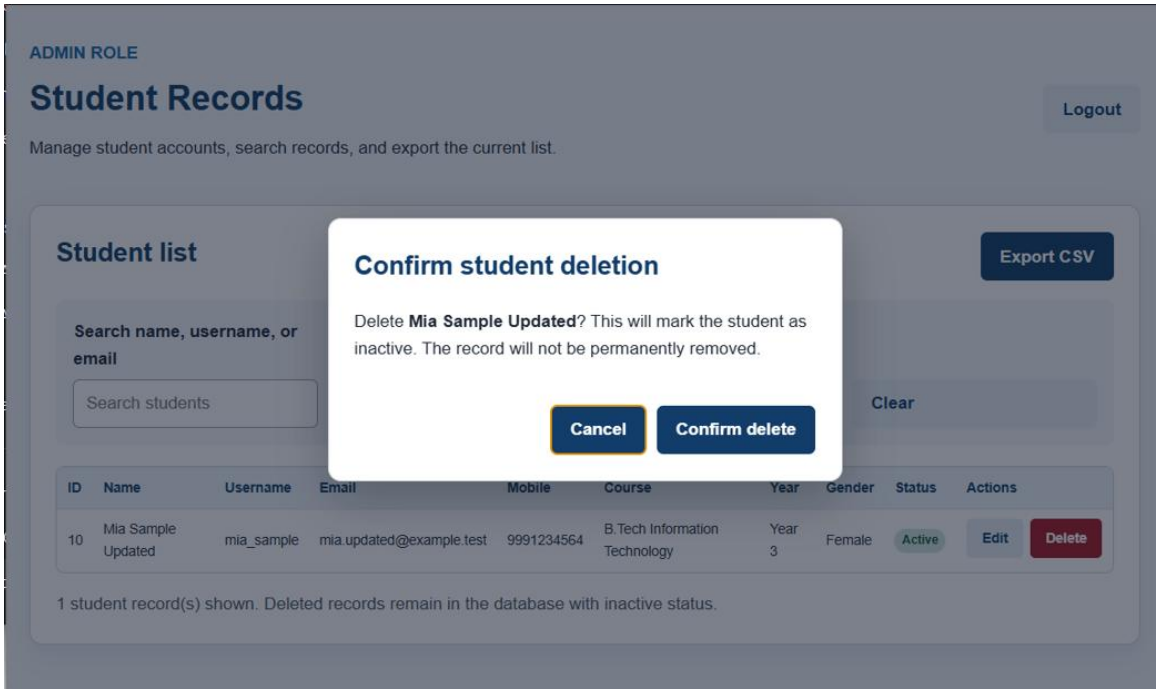
3.4 Search functionality



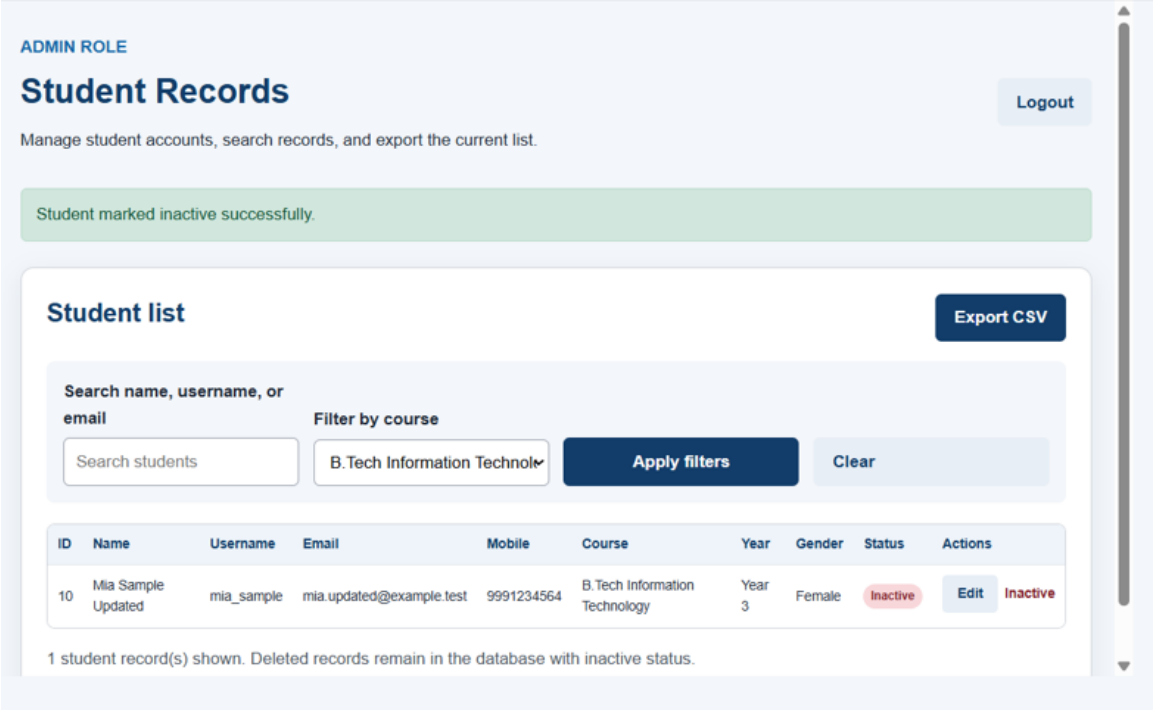
3.5 Course-wise filtering



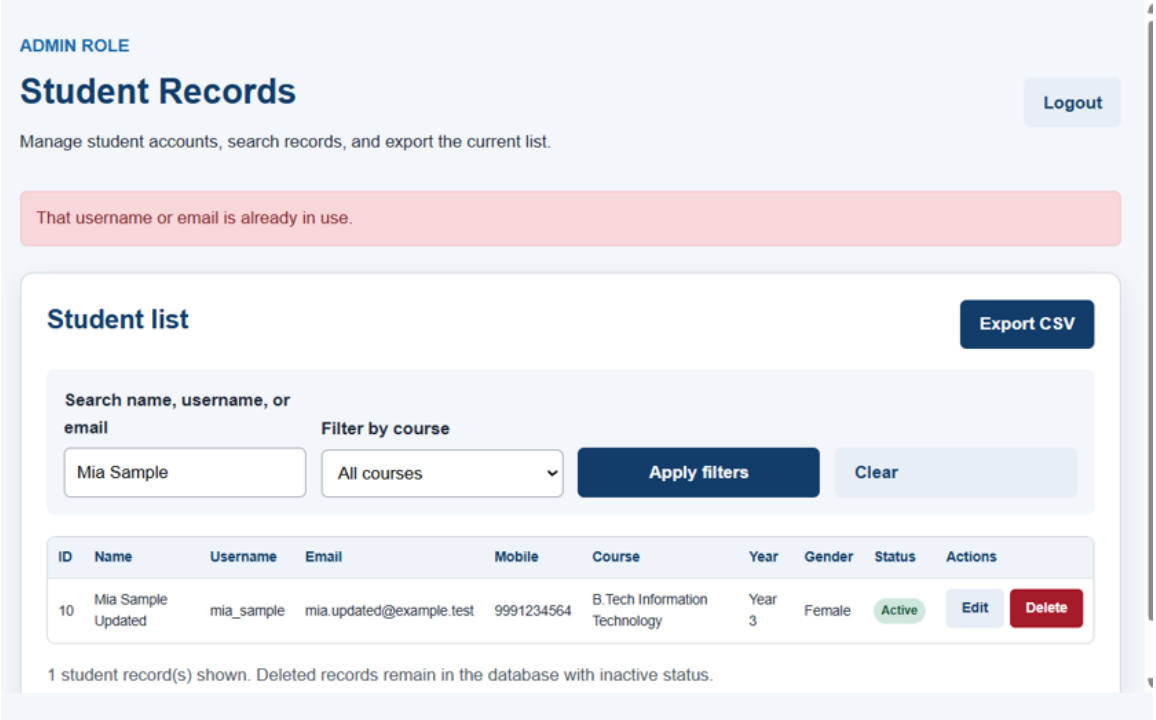
3.6 Delete confirmation message



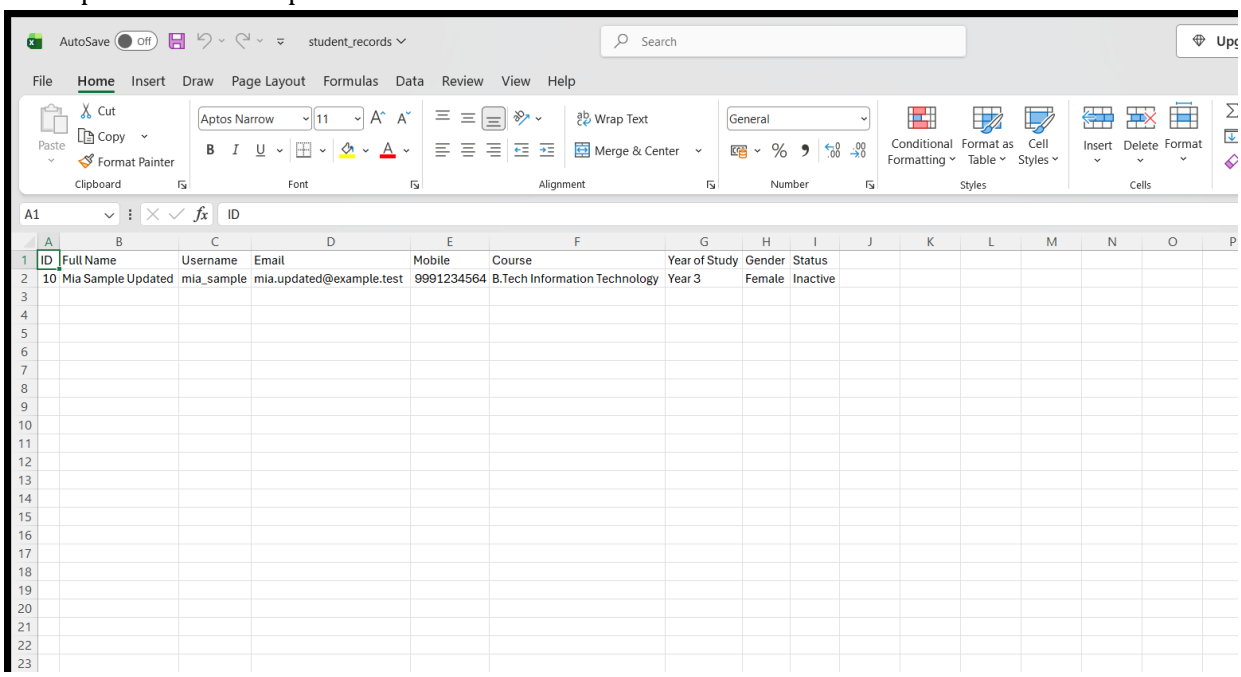
3.7 Successful soft delete (student status changed to inactive)



3.8 Success/error messages



3.9 Exported CSV file opened in Excel



ID	Full Name	Username	Email	Mobile	Course	Year of Study	Gender	Status
10	Mia Sample Updated	mia_sample	mia.updated@example.test	9991234564	B.Tech Information Technology	Year 3	Female	Inactive

4. Result

The advanced CRUD application was successfully implemented. Users can effectively view, add, update, and soft-delete student records. The search and filter functionalities correctly narrow down the displayed data, user actions are confirmed via dynamic success/error messages, and the export feature successfully generates a formatted CSV file of the records.

5. Conclusion

This practical reinforces the understanding of advanced PHP and MySQL database interactions. Implementing soft deletes preserves data integrity and history, while adding search, filter, and CSV export functionalities drastically improves the usability, data management, and real-world applicability of the web application.

— End of Practical 11 Submission —